Planning memo: 4-week revenue forecasts for a US Walmart store group

To: merchandise and finance planning. Data: M5 (10 stores in CA, TX, WI; 2011 to 2016). Test window: 6 rolling origins covering weeks 254 to 277 (2015-12-05 to 2016-05-14). Dollar figures below are in percent of revenue; parameters marked ASSUMED are scenario inputs, not estimates from the data.

Recommendation

- **Use the MinT-reconciled ARIMA forecast for store, category and department plans.** It has the lowest WAPE at every level below Total, and at Total it is within about 0.1 percentage points of the best model (SARIMA(1,1,1)(0,1,0)[52] is lowest, WAPE 2.56%). One coherent set of numbers means store plans add up to the company plan.
- **Do not use the 4-week moving average as the headline model, but keep it as the benchmark.** It beats ETS in WAPE at all 9 levels and is the hardest baseline to beat at store x department, yet it is 2 to 3 points worse than ARIMA at the top. Seasonal naive (WAPE 14% at Total) should not be used.
- **Size safety buffers from calibrated quantiles, not from the reconciled fable intervals.** Those intervals under-cover (for fable ARIMA, 80% intervals contain 17% to 71% of actuals at Total depending on the method). Among models with intervals, split-conformal LightGBM is closest to nominal at several of the lower levels.
- **Treat all model rankings among the top three as tentative:** there are only 6 test origins.

How accurate are the forecasts?

Model (WAPE %, test)	Total	State	Store	Department	Store x Dept
4-wk moving avg	5.0	5.3	5.8	6.5	8.1
SARIMA airline	2.6	3.3	4.1	4.4	6.7
LightGBM conf.	3.7	4.0	4.6	5.0	6.7
fable ARIMA+MinT	2.7	3.2	4.0	4.0	6.3

WAPE = sum of absolute errors divided by sum of actuals, pooled over 4-week-ahead forecasts of all nodes at the level (lower is better). Errors grow as the level gets more granular: the best store x department WAPE is several times the best Total WAPE.

How decision-useful are the intervals?

Model (80% interval coverage, %)	Total	State	Store	Department	Store x Dept
4-wk moving avg	62	62	69	67	71
SARIMA airline	75	74	75	76	75
LightGBM conf.	62	69	74	77	77
fable ARIMA+MinT	71	61	68	65	69

Nominal coverage is 80. Coverage at Total is estimated from 24 forecasts and is noisy. All models are below nominal at most levels.

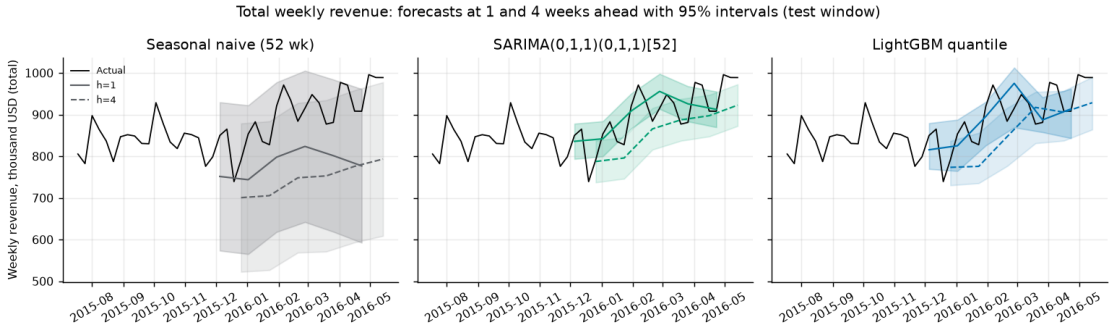
What does a safety buffer cost and save? (ASSUMED parameters)

Scenario: each dollar of revenue the plan misses costs 25 cents of margin (ASSUMED), and each dollar planned but not sold costs 5 cents of carrying and markdown cost (ASSUMED). Plan A orders at the point forecast. Plan B orders at the 90th percentile forecast (ASSUMED service level). Costs are summed over the nodes at each level and the 24 test weeks, shown as percent of actual revenue; the table `decision_sensitivity.csv` varies margin (15, 25, 35 percent) and carrying cost (2, 5, 10 percent).

Level	Model	Cost, point plan (% of revenue)	Cost, q90 plan (% of revenue)	Buffer, q90 over point (% of revenue)
Total	LightGBM conf.	0.78	0.23	4.3
Total	4-wk moving avg	1.04	0.39	5.8
Total	fable ARIMA+MinT	0.58	0.2	3.1
Total	SARIMA airline	0.53	0.19	3.3
Store x Dept	LightGBM conf.	1.27	0.75	12.3
Store x Dept	4-wk moving avg	1.5	0.77	10.9
Store x Dept	fable ARIMA+MinT	1.13	0.62	8.0
Store x Dept	SARIMA airline	1.18	0.63	9.7

Under these assumptions Plan B lowers the cost in every row shown, because missing a sale costs five times as much as carrying a dollar of excess. The buffer is small at Total and grows with granularity, since errors are larger and do not net out. Across all 1,215 model x level x scenario rows in the sensitivity table Plan B is cheaper in 98.5%; it loses only for margin 15% with carrying cost 10% (18 rows).

Realised service. The share of weeks in which actual revenue was at or below the q90 plan is exported per model, node and level in `decision_planning.csv` (column `share_weeks_actual_within_q90`). It is a check on whether q90 behaves like a 90th percentile; for several models it falls short of 0.90 (for example MinT ARIMA at Total, 0.75), so the q90 plan did not deliver a 90 percent service level in this window.



Limits

- M5 is 2011 to 2016 data from 10 stores in 3 states; the hierarchy, seasonality and SNAP effects may not hold elsewhere. Revenue is units x list price: promotions and stockouts are not separable from zero demand.
- Only 6 origins of 4 weeks form the final test; the planning table uses the same weeks, so it is an ex post illustration, not a forecast of savings.
- The item layer (top 300 items, top 600 item x store series) is a revenue-ranked sample, not reconciled; item x store WAPE is about 26% to 29% even for the best models.
- Residual autocorrelation remains in the ARIMA and ETS fits (Ljung-Box rejects at most nodes), so Gaussian intervals are approximate.